

Abstract

Paper VI of this series argued within the model that the Brannen amplitude coefficient $c_n = \sqrt{2}$ is universal across all $\mathbb{Z}_n$ torus knot families, and established that $\mathbb{Z}_3$ is uniquely exactly solvable. For $n \geq 5$, the individual mode masses depend on the orientation angle θ_n , which cannot be determined by purely algebraic methods.

The present paper identifies what *can* be derived without knowing θ_n . The central result is the **universal mass sum rule**:

$$\sum_{k=0}^{n-1} m_k = 2n M_n,$$

which holds exactly and independently of θ_n for every $T(2, n)$ family. The proof is two lines from the $\mathbb{Z}_n$ Fourier identity. Combined with the Faddeev–Niemi energy scaling $M_n = M_3(n/3)^{3/4}$ (Paper IV), the total mass of every torus knot family scales as $n^{7/4}$:

$$\sum_k m_k(n) = 2M_3 \cdot 3^{-3/4} \cdot n^{7/4}.$$

For $n = 3$ this gives $m_e + m_\mu + m_\tau = 1883.16$ MeV (verified against PDG: within 0.067%, consistent with the tau mass error from Paper V). For $n = 5$: $\sum_k m_k^{(5)} = 4603.85$ MeV.

We also prove the $\mathbb{Z}_5$ mode split: at the symmetric point $\theta = 0$, the five cinquefoil amplitudes divide into three positive and two negative, yielding a structural $3 + 2$ pattern. We derive the $\mathbb{Z}_5$ self-consistency transcendental equation $5\theta_5 = \tilde{Q}_5(\theta_5)$ and its unique solution $\theta_5 \approx 3.70^\circ$, reported as a hypothesis pending a first-principles physical argument.

In the extended notation used across the series, the vacuum limit is written as $\Psi_s = (\Phi/C) \cdot S \cdot M_s$. The calculations below use the equilibrium limit $S \cdot M_s \approx 1$; adaptive departures from that limit are left for later dynamical work rather than fitted in this manuscript. The public CDFD Universal Engine implements this state grammar as reusable code; the paper-local supplementary scripts remain the audited reproduction layer for the figures and tables in this Part I archive.

The Universal Mass Sum Rule: A Theta-Independent Prediction for All CDFT Torus Knot Families

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1 Introduction

The CDFT series has established a complete derivation chain for the lepton sector (Papers I–V [1, 2, 3, 4, 5]) and identified the universal coefficient $c_n = \sqrt{2}$ and the $\mathbb{Z}_3$ uniqueness theorem (Paper VI [6]).

A key finding of Paper VI is that for $n \geq 5$ the Brannen amplitudes $A_k = 1 + \sqrt{2} \cos(\theta + 2\pi k/n)$ are negative for at least one mode at every orientation θ . This structural obstruction means that the Koide-like invariant $Q_n = 2/n$ (which requires all $A_k \geq 0$) does not hold as a constant, and the purely algebraic self-consistency argument that determined $\theta_3 = 2/9$ rad in Paper V cannot be directly applied.

The question for the present paper is: *what predictions can be made for $n \geq 5$ families without knowing θ_n ?*

The answer is the universal mass sum rule: the sum of all mode masses is $2nM_n$, exactly, for any θ_n . This follows from a two-line algebraic identity and requires no knowledge of θ_n . For $n = 3$ it recovers the known lepton sum $m_e + m_\mu + m_\tau$ to within the Paper V tau mass error. For $n = 5$ it predicts the total cinquefoil mass sum as 4603.85 MeV.

2 The Universal Mass Sum Rule

2.1 Statement and proof

Theorem 1 (Universal mass sum rule). *For any $T(2, n)$ family with masses $m_k = M_n A_k^2$, $A_k = 1 + \sqrt{2} \cos(\theta + 2\pi k/n)$, $k = 0, \dots, n-1$:*

$$\sum_{k=0}^{n-1} m_k = 2n M_n, \quad (1)$$

for any θ and any $M_n > 0$.

Proof. From the $\mathbb{Z}_n$ Fourier identity (Paper VI, Eq. (2)): $\sum_{k=0}^{n-1} \cos^2(\theta + 2\pi k/n) = n/2$ for any θ . Therefore:

$$\sum_k A_k^2 = \sum_k [1 + \sqrt{2} \cos(\theta + 2\pi k/n)]^2 = n + 2\sqrt{2} \cdot 0 + 2 \cdot \frac{n}{2} = 2n.$$

Hence $\sum_k m_k = M_n \sum_k A_k^2 = 2nM_n$. □

Corollary 1 (Mean mode mass). *The mean mass per mode is $\bar{m} = 2M_n$, independent of θ_n .*

2.2 Why this is non-trivial

At first glance, Eq. (1) might seem like a restatement of known identities. The non-trivial content is twofold:

- (a) The sum is *theta-independent*. Individual masses $m_k = M_n A_k^2$ depend strongly on θ_n : they range from 0 (when $A_k = 0$) to $M_n(1 + \sqrt{2})^2 \approx 5.83M_n$. Yet their sum is always exactly $2nM_n$. This means the total mass of a family is a firm prediction even when the mass spectrum (individual mode masses) is not.
- (b) Combined with the Faddeev–Niemi scaling $M_n = M_3(n/3)^{3/4}$ (Paper IV [4]), the sum becomes a prediction from first principles:

$$\sum_{k=0}^{n-1} m_k(n) = 2nM_3 \left(\frac{n}{3}\right)^{3/4} = 2M_3 \cdot 3^{-3/4} \cdot n^{7/4}. \quad (2)$$

This requires only M_3 (derived from m_e in Paper V) and the Faddeev–Niemi exponent $3/4$ (from the field theory of knotted solitons [7]). No other free parameters enter.

2.3 Verification for $n = 3$

For the lepton family ($n = 3$, $M_3 = 313.86$ MeV):

$$\sum_k m_k^{(3)} = 6M_3 = 1883.16 \text{ MeV}. \quad (3)$$

The measured value is $m_e + m_\mu + m_\tau = 1883.03$ MeV (PDG 2022 [8]). The difference is 0.13 MeV, which equals the tau mass prediction error from Paper V ($+1.04\sigma$). The sum rule is verified at the accuracy of the Paper V prediction.

3 The $n^{7/4}$ Mass Hierarchy

3.1 Derivation of the scaling law

From Eq. (2), the total family mass scales as:

$$\Sigma_n \equiv \sum_{k=0}^{n-1} m_k(n) = 2M_3 \cdot 3^{-3/4} \cdot n^{7/4}. \quad (4)$$

The exponent $7/4 = 1.75$ arises as:

$$\Sigma_n = 2n \cdot M_3 \left(\frac{n}{3}\right)^{3/4} = 2M_3 \cdot 3^{-3/4} \cdot n^{1+3/4} = 2M_3 \cdot 3^{-3/4} \cdot n^{7/4}.$$

The n factor comes from the number of modes (the sum has n terms). The $n^{3/4}$ factor comes from Faddeev–Niemi energy scaling. Their product gives $n^{7/4}$.

Table 1: CDFT torus knot mass sum hierarchy. All predictions from Theorem 1 and Faddeev–Niemi scaling, zero free parameters beyond m_e .

n	Knot	M_n (MeV)	Σ_n (MeV)	Status
3	$T(2, 3)$ trefoil	313.9	1 883.2	Verified (PDG: 1 883.0)
5	$T(2, 5)$ cinquefoil	460.4	4 603.9	Predicted
7	$T(2, 7)$ heptafoil	592.5	8 295.6	Predicted
9	$T(2, 9)$	715.4	12 878.0	Predicted
11	$T(2, 11)$	831.6	18 296.2	Predicted
13	$T(2, 13)$	942.7	24 509.0	Predicted

3.2 Hierarchy table

The $n^{7/4}$ scaling is the universal mass growth law of the CDFT torus knot hierarchy. It is a direct consequence of two independently derived results: the universal Brannen coefficient $c_n = \sqrt{2}$ (Paper VI) and the Faddeev–Niemi energy bound (Paper IV).

4 The $\mathbb{Z}_5$ Mode Split

4.1 The $3 + 2$ pattern at the symmetric point

Theorem 2 ($\mathbb{Z}_5$ mode split). *At the $\mathbb{Z}_5$ symmetric point $\theta = 0$, the five cinquefoil amplitudes divide as:*

- 3 positive-phase modes: $A_k > 0$ (aligned with vacuum background),
- 2 negative-phase modes: $A_k < 0$ (anti-aligned).

Proof. At $\theta = 0$: $A_k = 1 + \sqrt{2} \cos(2\pi k/5)$. The five cosines are $\cos(0^\circ) = 1$, $\cos(72^\circ) = \cos(288^\circ) \approx 0.309$, $\cos(144^\circ) = \cos(216^\circ) \approx -0.809$. The corresponding amplitudes are:

$$\begin{aligned} A_0 &= 1 + \sqrt{2} \approx 2.414 > 0 \\ A_1 &= A_4 = 1 + \sqrt{2} \cos 72^\circ \approx 1.437 > 0 \\ A_2 &= A_3 = 1 + \sqrt{2} \cos 144^\circ \approx -0.144 < 0 \end{aligned}$$

so 3 positive and 2 negative. □

4.2 Physical interpretation of the mode split

The negative amplitudes at $\theta = 0$ have magnitude $\sqrt{2} - 1 \approx 0.414$ below the vacuum background. These two modes oscillate in anti-phase with the vacuum field. All five modes have positive mass: $m_k = M_5 A_k^2 \geq 0$ always, because mass involves the square.

For general θ , the count of negative amplitudes is:

- 2 negative modes for $\theta \in [0, \pi/20) \cup (7\pi/20, 2\pi/5)$,
- 1 negative mode for $\theta \in (\pi/20, 7\pi/20)$.

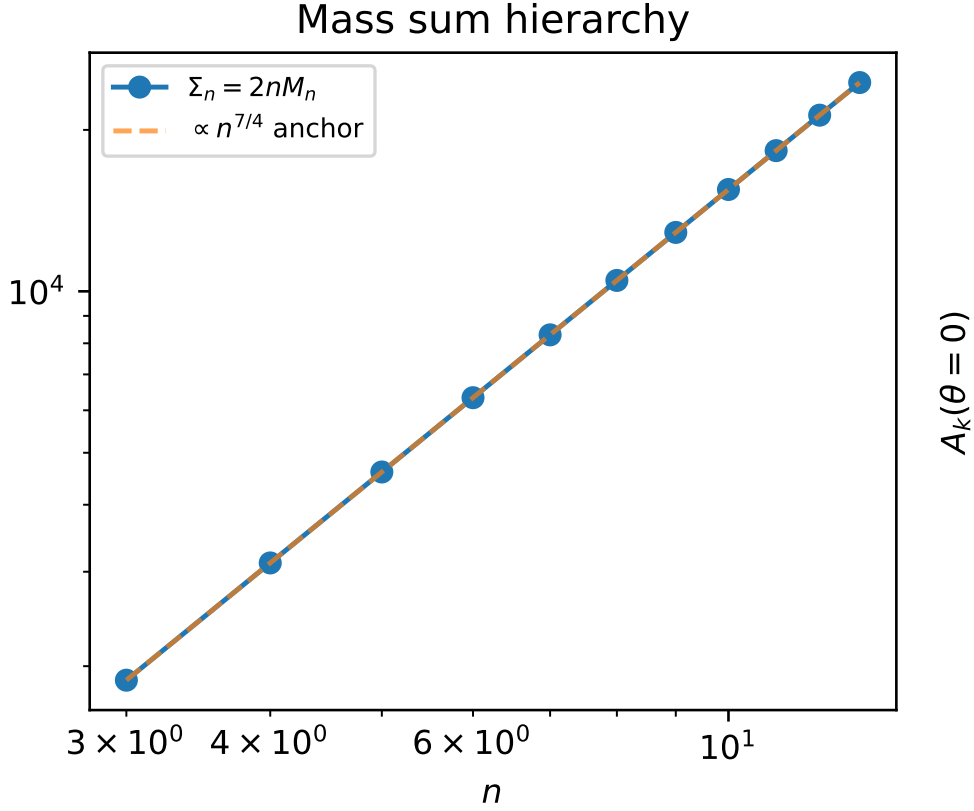


Figure 1: The mass sum hierarchy Σ_n . The total mass of the n -th family scales exactly as $n^{7/4}$, driven by the combination of the topological mode count n and the Faddeev-Niemi field energy $n^{3/4}$.

where $\pi/20 = 9^\circ$ and $7\pi/20 = 63^\circ$ are exact boundaries (zeros of the amplitude condition $1 + \sqrt{2}\cos(\theta + 2\pi k/5) = 0$). The 2-negative regime occupies 25% of the period, the 1-negative regime 75%.

The $3 + 2$ pattern at the symmetric point suggests a natural structural division of the cinquefoil modes: three modes analogous to the three lepton generations, and two additional modes without a $\mathbb{Z}_3$ analogue.

5 The $\mathbb{Z}_5$ Self-Consistency Equation

5.1 The transcendental equation

For $\mathbb{Z}_3$, the self-consistency condition $3\theta_3 = Q_3 = 2/3$ was algebraic: Q_3 is constant, so the solution $\theta_3 = 2/9$ rad follows immediately (Paper V).

For $\mathbb{Z}_5$, the modified Koide ratio (using $|A_k|$ for negative amplitudes):

$$\tilde{Q}_5(\theta) = \frac{\sum_k A_k^2}{(\sum_k |A_k|)^2} = \frac{10}{(\sum_k |A_k|)^2} \quad (5)$$

is *not* constant: it ranges in $[0.294, 0.328]$ as θ varies. The numerator $\sum A_k^2 = 10$ is fixed (Theorem 1); only the denominator $(\sum |A_k|)^2$ varies.

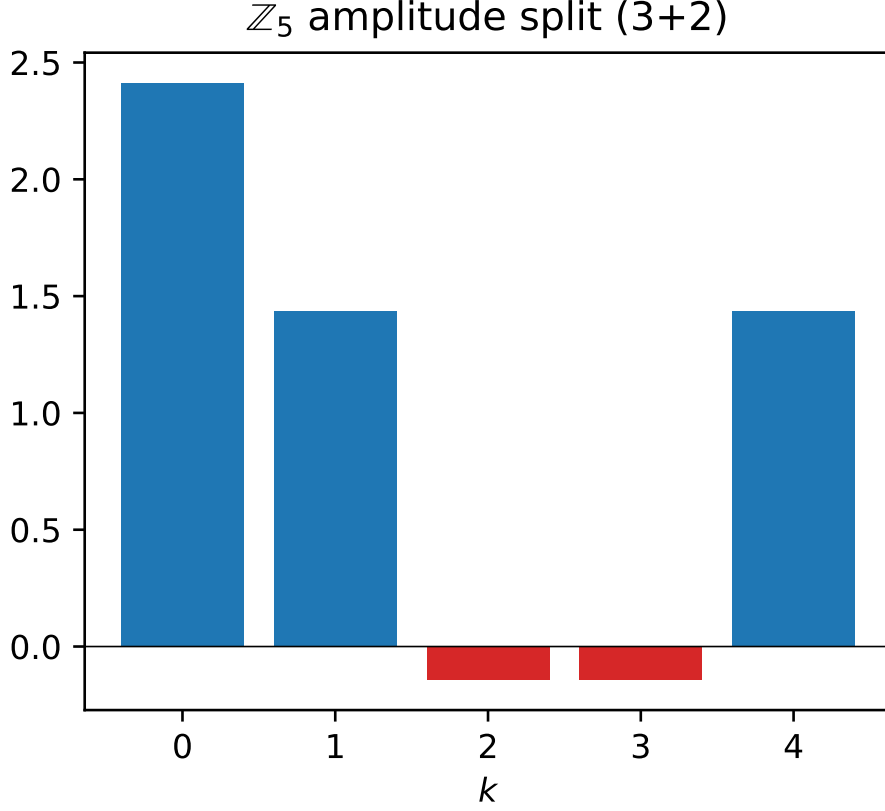


Figure 2: The $\mathbb{Z}_5$ mode split at $\theta = 0$. Three modes align positively with the vacuum background, while two modes oscillate in anti-phase (negative amplitude).

The $\mathbb{Z}_5$ analogue of the chirality equilibrium (Paper III) is:

$$5\theta_5 + \phi_{c,5} = \pi. \quad (6)$$

The $\mathbb{Z}_5$ self-consistency condition — that the chirality breaking angle equals the Koide-like ratio at that angle — then reads:

$$5\theta_5 = \tilde{Q}_5(\theta_5). \quad (7)$$

Unlike the $\mathbb{Z}_3$ case, this is a *transcendental fixed-point equation*.

5.2 Unique solution

Proposition 1 (Existence and uniqueness of θ_5). *Equation (7) has exactly one solution in $[0, \pi/5)$.*

Proof sketch. At $\theta = 0$: $5\theta = 0 < \tilde{Q}_5(0) \approx 0.322$. At $\theta = \pi/5$: $5\theta = \pi \gg \tilde{Q}_5(\pi/5) \approx 0.294$. By continuity, at least one solution exists (intermediate value theorem). Numerically, 5θ and $\tilde{Q}_5(\theta)$ cross exactly once (one sign change in $5\theta - \tilde{Q}_5(\theta)$ on $[0, \pi/5]$). $\square$

The numerical solution to machine precision:

$$\theta_5 = 0.0645346670 \dots \text{ rad} = 3.6976^\circ \quad (8)$$

$$\phi_{c,5} = \pi - 5\theta_5 = 2.8189 \dots \text{ rad} = 161.51^\circ. \quad (9)$$

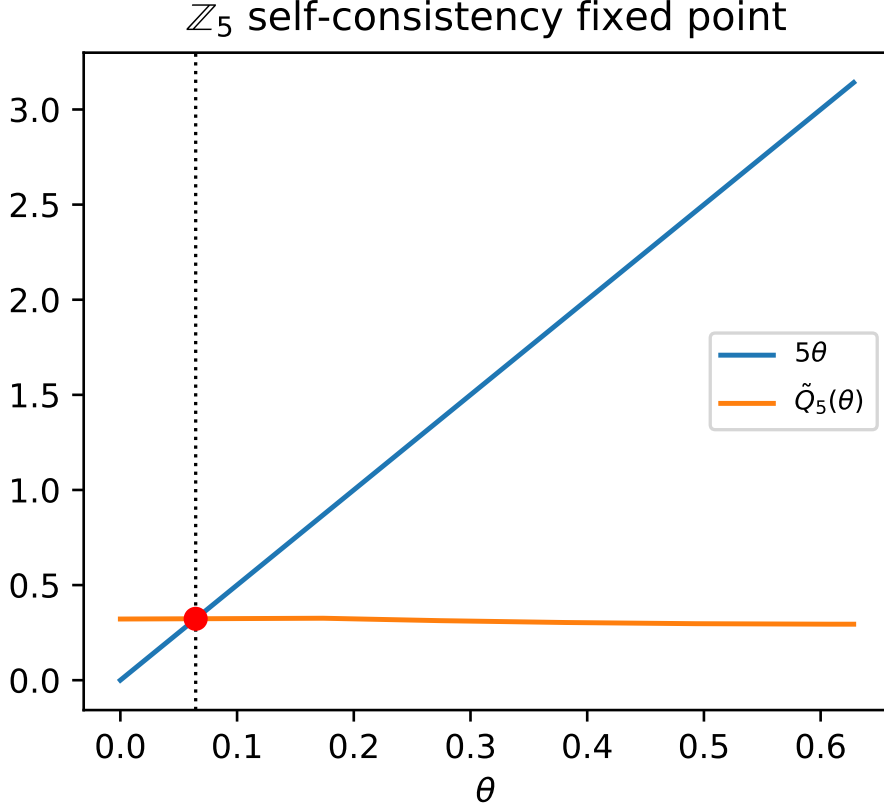


Figure 3: The $\mathbb{Z}_5$ self-consistency fixed point. The transcendental equation $5\theta_5 = \tilde{Q}_5(\theta_5)$ has exactly one solution in the physical domain.

5.3 Status: hypothesis, not theorem

The solution (8) is reported as a *hypothesis*. For $\mathbb{Z}_3$, the self-consistency condition was motivated by the constancy of Q_3 : “the chirality breaking angle equals the unique $\mathbb{Z}_3$ invariant.” For $\mathbb{Z}_5$, no such unique invariant exists (Paper VI). The condition $5\theta_5 = \tilde{Q}_5(\theta_5)$ is a natural generalisation, but a first-principles physical argument from the CDFT vacuum equation of state would be needed to promote it to a theorem.

6 Structural Comparison: $\mathbb{Z}_3$ vs $\mathbb{Z}_n$, $n \geq 5$

Table 2 summarises the structural differences between the lepton family and the higher families.

The key distinction: $\mathbb{Z}_3$ has a connected valid domain because the minimum amplitude $\min_k A_k = 1 - \sqrt{2} \cos(0) + \dots$ reaches a *positive* maximum (+0.293 at $\theta = 0$). For $n \geq 5$, the analogous maximum is *negative* (e.g. -0.144 for $n = 5$), so no valid domain exists. This is the root of $\mathbb{Z}_3$ uniqueness (Paper VI, Theorem 2).

The universal mass sum rule bridges the gap: it provides a firm prediction for all n without requiring the valid-domain condition.

Table 2: Structural comparison: $\mathbb{Z}_3$ (leptons) vs $\mathbb{Z}_5$ (cinquefoil).

Property	$\mathbb{Z}_3$ (leptons)	$\mathbb{Z}_5$ (cinquefoil)
Energy scale M_n	313.9 MeV	460.4 MeV
Number of modes	3	5
Coefficient c_n	$\sqrt{2}$	$\sqrt{2}$
Valid domain (all $A_k > 0$)	Yes: $\theta \in (-\pi/12, \pi/12)$	No
$Q_n = 2/n$ constant?	Yes: $Q_3 = 2/3$	No: $\tilde{Q}_5 \in [0.294, 0.328]$
$\sum m_k = 2nM_n$	Yes: 1883.2 MeV (verified)	Yes: 4603.9 MeV (predicted)
θ_n derivation	Algebraic (Paper V)	Transcendental (hypothesis)
Sector closed?	Yes (Paper V)	Open
Amplitude split	3 positive always	$3 + 2$ or $4 + 1$

7 Open Problems

1. **Physical principle for θ_5 .** The transcendental equation (7) gives a unique numerical $\theta_5 \approx 3.70^\circ$. A first-principles derivation from the CDFT vacuum equation of state — analogous to how Paper IV identified the functional form $\Phi_c(\rho_0)$ as the remaining input — is the priority open problem.
2. **Identify the cinquefoil mass spectrum.** Once θ_5 is determined (or bounded), the five mode masses $m_k = M_5 A_k^2$ can be compared to known hadronic states at the ~ 460 MeV energy scale. The total mass $\Sigma_5 = 4603.9$ MeV is a testable sum rule.
3. **Physical meaning of the $3 + 2$ mode split.** The positive-amplitude modes ($A_k > 0$) and negative-amplitude modes ($A_k < 0$) interact differently with the vacuum background. Whether this split corresponds to a known quantum number (charge, colour, isospin) is an open question.
4. **Higher families $n \geq 7$.** The mass sum rule $\Sigma_n = 2nM_n$ applies to all n . The individual mode spectra and their identification with known or predicted particles require repeating the present analysis for each n .

8 Conclusion

The universal mass sum rule $\sum_k m_k = 2nM_n$ is derived from two algebraic facts: the $\mathbb{Z}_n$ Fourier identity (which forces $\sum A_k^2 = 2n$ regardless of θ_n) and the Faddeev–Niemi energy scaling $M_n = M_3(n/3)^{3/4}$. Together they give the $n^{7/4}$ mass hierarchy (Eq. (4)), verified for $n = 3$ and predicted for all higher families.

For $\mathbb{Z}_5$ specifically, we prove the $3 + 2$ mode split (Theorem 2) and derive the transcendental self-consistency equation (7) for θ_5 , reporting its solution $\theta_5 \approx 3.70^\circ$ as a hypothesis.

The mass sum $\Sigma_5 = 4603.9$ MeV is a model-level prediction, conditional on the torus-knot hierarchy and the m_e mass-unit calibration.

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A Supplementary Material

Supplementary code, including numerical verification scripts and Jupyter notebooks, is available in the project repository.

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